COHORT 17 — AI PROJECT & PRD ASSIGNMENT

UIRevive — AI Website UI Improvement Platform

Product Requirements Document (PRD) & Assignment Submission

Student Name:	Muhammad Osama Khan	**Cohort:**	Cohort 17
Project Title:	UIRevive (AI-Powered Website UI Improvement Platform)	**Submission Date:**	November 1, 2026

1. Project Idea & Overview

Project Name: UIRevive (AI-powered Website UI Improvement Platform)
Core Idea: A platform where users enter the URL of an existing website and receive an understandable UI improvement report. The platform aims to identify design weaknesses such as poor visual hierarchy, inconsistent colors, typography problems, spacing issues, and unclear layouts. It suggests practical improvements while respecting the website's existing brand identity, instead of making every website look like the same AI-generated template.

2. Problem Statement

Many websites have UI problems such as inconsistent spacing, weak visual hierarchy, poor typography, confusing layouts, and inconsistent colors. Website owners and developers often recognize that their site needs improvement but may not know exactly what to change or why. At the same time, existing AI-generated redesign tools output homogenized, cookie-cutter templates that discard original branding. UIRevive presents structured, actionable critique paired with a before-and-after concept while preserving unique brand essence.

3. Why I Chose This Idea

- **Target Audience:** Solves practical challenges for freelance web developers, UI/UX designers, web agencies, and small business owners.
- **Brand-Respecting:** Preserves existing visual identity rather than generating generic AI template designs.
- **Clean Engineering:** Next.js frontend prototype built with React, TypeScript, and Tailwind CSS, backed by a resilient server-side AI integration route with mock fallback.

4. Structured Project JSON Specification


```
{
  "Idea": "UIRevive — AI-powered Website UI Improvement Platform",
  "Statement": "A platform where users enter the URL of an existing website and receive an understandable UI impr
ovement report. The platform aims to identify design weaknesses such as poor visual hierarchy, inconsistent color
s, typography problems, spacing issues, and unclear layouts. It should suggest practical improvements while respe
cting the website's existing brand identity, instead of making every website look like the same AI-generated temp
late.",
  "Coding environment": "AI code editor using Next.js, React, TypeScript, and Tailwind CSS",
  "Current development scope": "Frontend plus one server-side API route that calls Ollama. Use mock data as a fal
lback. Do not create a database, separate server, or website crawler in this phase.",
  "API Key": "Ollama API key provided by the user, stored in OLLAMA_API_KEY in .env.local (server-side only, neve
r committed or exposed to the browser).",
  "Features": [
    "Landing page explaining the platform",
    "Website URL input with basic frontend validation",
    "Analyze Website button",
    "Animated simulated scanning experience",
```

```
    "Demo analysis report with an overall UI score",
    "Sample analysis categories: visual hierarchy, typography, color consistency, spacing and alignment, layout c
larity, and responsiveness",
    "UI issue cards showing the problem, reason, severity, and recommendation",
    "Before-and-after example of an improved website section",
    "Responsive navigation and layouts"
  ],
  "Themes": "Use a modern, professional color system. Provide a cohesive light theme as the default. A dark theme
can be included if it does not delay the core prototype.",
  "Style": "Minimalistic, sleek, modern SaaS interface with subtle animations and smooth transitions. Avoid gener
ic AI-dashboard styling and excessive visual effects.",
  "Responsiveness": "Fully responsive across mobile, tablet, laptop, and desktop.",
  "Look and feel": "Modern, polished, professional, distinctive, and easy to understand.",
  "Layout": "User-friendly layout with a clear landing page, simple URL submission flow, readable analysis repor
t, and easy-to-compare improvement preview.",
  "Design": "Clean typography, strong visual hierarchy, consistent spacing, reusable components, and restrained u
se of color.",
  "Interface": "Next.js web application built with React, TypeScript, and Tailwind CSS.",
  "User workflow": "Open the landing page, enter a website URL, click Analyze Website, view a simulated scanning
animation, review the demo report, explore recommendations, and view a before-and-after design example.",
  "Data handling": "The entered URL is sent only to the /api/analyze route, which forwards it to Ollama. Nothing
is stored. Local mock data is used as a fallback.",
  "Important limitation": "The current version does not fetch or scan the submitted website. Ollama generates sco
res, issues, and recommendations from the URL alone, so they are AI estimates, not measurements.",
  "Future scope": "Add real website screenshot capture, website analysis, vision-capable AI integration, brand-aw
are recommendations, and generated design previews after the frontend prototype is complete."
}
```

5. How It Works & User Journey

1. The user opens the UIRevive landing page at https://ui-revive.vercel.app/.
2. The user pastes a website URL into the input field (with client-side validation).
3. The user clicks **Analyze Website**.
4. The application displays an animated scanning experience simulating design checks.
5. The server-side route (`/api/analyze`) requests a structured design critique from the AI model (with mock data fallback).
6. The application opens the analysis report displaying overall score, 6 category breakdowns, and filterable issues (High, Medium, Low severity).
7. The user views a Before-and-After improvement preview demonstrating practical refinements.

6. AI Model, API Provider & Security

- **Model & Provider:** Multi-provider server route supporting Ollama, Groq (Llama 3.3 70B), and Google Gemini (Gemini 1.5 Flash).
- **Data Handling:** The entered URL is passed to the server-side API only. Nothing is stored in a database.
- **Security:** All API keys are loaded server-side through `.env.local`. Never exposed to browser bundles or committed to git (configured in `.gitignore`).

7. Project Submission Links

Requirement	Submission Link / Details
Working Application Link	https://ui-revive.vercel.app/
GitHub Code Repository	https://github.com/OsamaBangash8840/UIRevive
PRD Document	Submitted via this PDF Document

8. Pre-Submission Checklist

- ✓ My project idea is useful, original, and solves a real problem for developers and businesses.
- ✓ My PRD includes all required points and technical constraints.
- ✓ I have included my full structured Project JSON specification.
- ✓ I have named the AI model and API provider integration.
- ✓ I have explained the complete user workflow steps.
- ✓ I have written, tested, and verified my code without errors.
- ✓ My working application link opens and runs live at https://ui-revive.vercel.app/.
- ✓ My code is committed and pushed to GitHub at https://github.com/OsamaBangash8840/UIRevive.
- ✓ My name (**Muhammad Osama Khan**) and **Cohort 17** are included.
- ✓ I have safely kept all API keys in environment variables without exposing them publicly.